

BLUESTOCK MUTUAL FUND ANALYTICS

An End-to-End Data Analytics Capstone Project

Mutual Fund Performance, Risk, and Investor Behaviour Analysis using Python, SQL, and Power BI

A Project Report Submitted in Partial Fulfilment of the Requirements for the

Degree of Bachelor of Technology

in Computer Science and Engineering (Artificial Intelligence & Data Science)

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Declaration

I hereby declare that the project work titled “Bluestock Mutual Fund Analytics”, submitted as part of my internship/capstone requirement, is an original piece of work carried out by me under the guidance of Yash Kale. All data sources, tools, and libraries used have been duly acknowledged, and the analysis, code, and dashboard design presented in this report represent my own independent effort.

I further declare that this report has not been submitted, in full or in part, to any other university, institute, or organisation for the award of any degree, diploma, or certification.

Place: Pune(remote)

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Executive Summary

The mutual fund industry in India has grown into a multi-trillion-rupee market, with hundreds of schemes spread across equity, debt, hybrid, and solution-oriented categories. For an investor, an advisor, or an analyst, comparing funds on the basis of returns alone is insufficient — real investment decisions require a joint view of returns, risk, cost, and investor behaviour. This project, Bluestock Mutual Fund Analytics, was undertaken to build exactly that joint view: an end-to-end analytics pipeline that ingests raw fund data, cleans and structures it, computes industry-standard performance and risk metrics, and presents the results through an interactive Power BI dashboard.

The project was executed over six working days, moving from raw CSV ingestion to an advanced analytics layer. Ten datasets covering fund master information, NAV history, AUM, SIP inflows, category-wise inflows, industry folio counts, scheme performance, investor transactions, portfolio holdings, and benchmark indices were consolidated into a relational SQLite database designed on a star schema. On top of this foundation, a performance analytics module computed daily returns, CAGR (1-year and 3-year), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown for all 40 mutual fund schemes in scope, producing a consolidated Fund Scorecard and ranking.

A four-page Power BI dashboard was then built to present Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends, complete with KPI cards, scatter plots, heatmaps, slicers, drill-through pages, and tooltips. The project was extended with an advanced analytics layer covering Value-at-Risk (95%), Conditional VaR, rolling 90-day Sharpe Ratio, investor cohort analysis, SIP continuity analysis, a rule-based fund recommender, and portfolio concentration analysis using the Herfindahl-Hirschman Index (HHI).

Key findings from the analysis show that the Mirae Asset Large Cap Fund recorded the highest risk-adjusted returns among the funds studied, registering the highest Sharpe Ratio and Sortino Ratio in the sample, while the SBI Small Cap Fund exhibited the highest Value-at-Risk, reflecting the higher volatility typically associated with small-cap exposure. These results are consistent with the broader risk-return trade-off expected across market-capitalisation categories and validate the analytical pipeline built in this project.

Overall, the project demonstrates a complete data analytics workflow — ingestion, cleaning, database design, exploratory analysis, quantitative performance and risk analytics, business intelligence dashboarding, and advanced statistical modelling — applied to a real-world financial domain, and is intended to serve both as a functional analytics tool and as a demonstration of applied data science and financial engineering skills.

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1. Introduction

Mutual funds pool money from a large number of investors and deploy it across equity, debt, and money-market instruments according to a stated investment objective. In India, the mutual fund industry is regulated by the Securities and Exchange Board of India (SEBI) and disseminated through the Association of Mutual Funds in India (AMFI), both of which publish standardised data such as daily Net Asset Values (NAVs), Assets Under Management (AUM), and folio counts. This standardisation makes mutual fund data an excellent domain for building an end-to-end data analytics project, since it combines time-series analysis, financial risk modelling, relational database design, and business intelligence visualisation in a single, industry-relevant context.

The Bluestock Mutual Fund Analytics project was conceived as a capstone/internship exercise to simulate the kind of analytics function that exists inside asset management companies (AMCs), wealth-management platforms, and fintech research teams. Rather than treating the problem as a single notebook exercise, the project was structured as a six-day pipeline moving from raw data ingestion to advanced quantitative analytics, mirroring how analytics teams incrementally build capability: first making the data trustworthy, then descriptive, then diagnostic, then predictive/advisory.

This report documents the complete lifecycle of the project — the technology choices made, the data engineering performed, the financial metrics computed, the dashboard designed, and the insights derived — and is organised so that a reader unfamiliar with the project can understand both what was built and why each design decision was made.

2. Problem Statement

Retail investors, financial advisors, and internal AMC analysts are frequently presented with mutual fund data in fragmented, siloed forms — NAV history in one file, AUM figures in another, transaction-level data in a third — with no unified system to relate fund performance to risk, cost, and investor behaviour simultaneously. Existing fact sheets published by AMCs typically report only point-in-time returns and a small set of ratios, without offering interactivity, category-level comparison, or investor-behaviour context such as SIP continuity or redemption patterns.

This creates three concrete problems that the project sets out to solve:

- **Fragmentation:** Fund master data, NAV history, AUM, SIP inflows, and transaction records exist as separate, unrelated files with no common analytical layer connecting them.
- **Shallow risk visibility:** Standard fact sheets emphasise returns but rarely present a fund's risk profile — volatility, drawdown, Value-at-Risk, or tail risk — in a way that is easy to compare across schemes.
- **Lack of interactivity:** Static PDF fact sheets and one-off spreadsheets cannot be sliced by fund house, category, plan type, investor state, or age group, which limits their usefulness for exploratory decision-making.

Bluestock Mutual Fund Analytics addresses these problems by building a single, structured data pipeline that ingests all relevant datasets, computes a comprehensive set of performance and risk metrics, and exposes the results through an interactive, multi-page Power BI dashboard supported by a Python-based analytics and recommendation layer.

3. Objectives

The project was guided by the following specific, measurable objectives:

- Design and implement a reliable ETL (Extract, Transform, Load) pipeline capable of ingesting ten heterogeneous mutual fund datasets into a single, query-ready relational database.
- Clean and validate the ingested data, addressing missing values, duplicate records, and inconsistent date formats to ensure analytical accuracy.
- Design a star-schema database in SQLite that supports efficient joins between fund, NAV, transaction, and benchmark data.
- Conduct exploratory data analysis (EDA) to understand NAV trends, AUM growth, SIP inflow patterns, and category-level dynamics.
- Compute industry-standard performance metrics — CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown — for all 40 mutual fund schemes in the dataset.
- Benchmark scheme performance against relevant market indices to assess relative performance.
- Build a four-page interactive Power BI dashboard covering industry overview, fund performance, investor analytics, and SIP/market trends.
- Extend the analysis with advanced risk and behavioural analytics: Historical VaR, CVaR, rolling Sharpe Ratio, investor cohort analysis, SIP continuity analysis, a rule-based recommender, and portfolio concentration (HHI) analysis.
- Present all findings in a structured, reproducible format suitable for both academic evaluation and portfolio demonstration.

4. Technology Stack

The project uses a combination of Python-based data engineering tools, a lightweight relational database, and Power BI for business intelligence, supported by Git/GitHub for version control. The table below summarises the stack and the role each tool plays in the pipeline.

Category	Tool / Library	Role in Project
Programming Language	Python 3	Core language for data ingestion, cleaning, and analytics
Data Manipulation	Pandas	Data wrangling, cleaning, aggregation, time-series handling
Numerical Computing	NumPy	Vectorised calculations for returns, ratios, and statistical measures
Static Visualisation	Matplotlib	Exploratory charts and the rolling Sharpe Ratio chart
Interactive Visualisation	Plotly	Interactive EDA charts (NAV trends, benchmark comparisons)
Database	SQLite	Lightweight relational database storing the cleaned star schema
ORM / DB Toolkit	SQLAlchemy	Python-to-SQL abstraction layer for querying and loading data
Business Intelligence	Power BI	Four-page interactive dashboard with slicers, drill-through, and tooltips
Version Control	Git	Local versioning of code and notebooks
Code Hosting	GitHub	Remote repository hosting and project documentation
Development Environment	Jupyter Notebook	Interactive development for EDA and analytics
Code Editor	VS Code	Primary editor for scripts, the recommender module, and project structure

5. Dataset Description

The project consolidates ten distinct datasets, each capturing a different dimension of the mutual fund ecosystem — from static fund attributes to time-varying NAV, flow, and transaction data. The table below describes each file and its analytical purpose.

#	File Name	Description / Purpose
1	01_fund_master.csv	Static reference data for each scheme: fund name, fund house, category, plan type, and launch date
2	02_nav_history.csv	Daily/periodic NAV time series used for return, CAGR, and drawdown calculations
3	03_aum_by_fund_house.csv	AUM figures aggregated by fund house, used for industry overview KPIs
4	04_monthly_sip_inflows.csv	Monthly SIP inflow values used for SIP trend and continuity analysis
5	05_category_inflows.csv	Net inflows/outflows broken down by fund category (equity, debt, hybrid, etc.)
6	06_industry_folio_count.csv	Folio (investor account) counts used as a proxy for investor participation
7	07_scheme_performance.csv	Pre-computed or supporting performance figures used to validate calculated metrics
8	08_investor_transactions.csv	Transaction-level records (SIP, lumpsum, redemption) with state and age-group fields
9	09_portfolio_holdings.csv	Underlying holdings per scheme, used for the HHI concentration analysis
10	10_benchmark_indices.csv	Benchmark index levels (e.g., NIFTY 50) used for Alpha/Beta and relative performance

Together, these datasets span 40 mutual fund schemes across multiple fund houses and categories, giving the analysis sufficient breadth to compare performance and risk both within and across categories.

6. Project Architecture

The system follows a layered architecture typical of modern analytics pipelines: a raw data layer, an ETL/cleaning layer, a structured storage layer, an analytics layer, and a presentation layer. Data flows one-directionally from raw CSV files through to the Power BI dashboard, with the SQLite database acting as the single source of truth consumed by both the Python analytics scripts and Power BI.

[Figure 1: End-to-End System Architecture of Bluestock Mutual Fund Analytics]

At a high level, the architecture consists of the following stages:

- Raw Data Layer: Ten CSV source files representing fund master data, NAV history, AUM, flows, transactions, holdings, and benchmarks.
- Ingestion Layer: Python/Pandas scripts read each CSV, with an additional live NAV-fetching routine to supplement historical data.
- Cleaning & Validation Layer: Missing value handling, duplicate removal, date standardisation, and schema validation.
- Storage Layer: A SQLite database designed on a star schema, loaded via SQLAlchemy.
- Analytics Layer: Python modules computing returns, risk-adjusted performance ratios, VaR/CVaR, rolling metrics, cohort analysis, and the rule-based recommender.
- Presentation Layer: A four-page Power BI dashboard connected to the SQLite database, plus exported CSV/PNG artefacts (e.g., `var_cvar_report.csv`, `rolling_sharpe_chart.png`).

This layered design keeps each stage independently testable: cleaning logic can be validated without touching analytics, and analytics can be validated without touching the dashboard, which is a good practice for maintainable data pipelines.

7. ETL Pipeline

The ETL (Extract, Transform, Load) pipeline forms the backbone of the project and was implemented as the first milestone (Day 1). It is responsible for reliably bringing all ten source datasets into a consistent, analysis-ready form.

7.1 Extract

All ten CSV files were read into Pandas DataFrames using a consistent ingestion routine that standardises column names, infers data types, and logs the row counts of each file for basic sanity checking. In addition to the static CSV extracts, a live NAV-fetching component was implemented to pull current-day NAV values, allowing the pipeline to be refreshed with near-real-time data rather than relying solely on historical snapshots.

7.2 Transform

The transformation stage standardises formats across all datasets — for example, aligning date columns to a single ISO format, harmonising fund identifiers between the fund master file and the NAV/transaction files, and converting monetary fields to a consistent numeric type. This stage also performs the data cleaning operations described in Section 8.

7.3 Load

Once cleaned, each DataFrame is loaded into the SQLite database using SQLAlchemy's engine and connection interface, with tables created according to the star schema described in Section 9. Loading is performed table-by-table with primary-key and foreign-key relationships enforced at the schema level so that downstream SQL queries and Power BI joins behave predictably.

A requirements.txt-driven environment setup (Day 1) ensured that all dependencies — pandas, numpy, sqlalchemy, matplotlib, plotly — were pinned to compatible versions, making the pipeline reproducible on a fresh machine.

8. Data Cleaning

Real-world financial datasets are rarely analysis-ready out of the box, and the ten source files used in this project were no exception. Day 2 of the project was dedicated to systematically cleaning and validating each dataset before it was loaded into the database.

8.1 Missing Value Handling

Missing values in NAV history were handled using forward-fill logic appropriate for time-series data, since a missing NAV on a non-trading day is best represented by the last known value rather than a zero or an average. Missing categorical fields in the fund master data (such as an unlabelled plan type) were flagged and, where possible, resolved using the scheme name; unresolvable records were excluded from ratio calculations to avoid skewing results.

8.2 Duplicate Removal

Duplicate rows — arising primarily from repeated exports in the transaction and NAV history files — were identified using composite keys (scheme code plus date, or transaction ID) and removed, retaining the first valid occurrence.

8.3 Date Formatting

Dates across the ten files arrived in multiple formats (DD-MM-YYYY, YYYY/MM/DD, and Excel serial dates in some exports). All date fields were parsed and standardised to the ISO 8601 (YYYY-MM-DD) format to ensure consistent time-series operations and correct chronological sorting.

8.4 Data Validation

After cleaning, each dataset was validated against basic business rules — for example, NAV values must be strictly positive, AUM figures must be non-negative, and transaction dates must fall within the overall NAV history range. Records failing validation were logged separately rather than silently dropped, preserving an audit trail of data quality issues.

9. Database Design

To support efficient, join-friendly querying from both Python and Power BI, the cleaned data was organised into a star schema within SQLite — a design pattern well suited to analytical (as opposed to transactional) workloads.

9.1 Star Schema Overview

The schema centres on two primary fact tables — NAV History and Investor Transactions — surrounded by dimension tables that describe funds, fund houses, categories, dates, and benchmark indices. This structure minimises data duplication while keeping the joins required for common analytical queries (e.g., “monthly SIP inflow by category”) shallow and fast.

Table	Type	Key Fields
Fund Master	Dimension	fund_id (PK), fund_name, fund_house, category, plan_type, launch_date
NAV History	Fact	fund_id (FK), date, nav
AUM by Fund House	Fact	fund_house (FK), date, aum
Monthly SIP Inflows	Fact	category (FK), month, sip_inflow
Category Inflows	Fact	category (FK), month, net_inflow
Industry Folio Count	Fact	date, category (FK), folio_count
Scheme Performance	Fact	fund_id (FK), period, return_pct
Investor Transactions	Fact	txn_id (PK), fund_id (FK), date, type, amount, state, age_group
Portfolio Holdings	Fact	fund_id (FK), stock_name, weight_pct
Benchmark Indices	Dimension/Fact	index_name, date, index_value

9.2 Design Rationale

SQLite was chosen over a heavier RDBMS for three reasons: it requires no separate server process, making the pipeline fully self-contained and portable; it integrates directly with Power BI's native database connector; and it is more than sufficient for the data volumes involved (tens of thousands of NAV and transaction rows, well within SQLite's practical limits). SQLAlchemy was used as the interface layer so that the same ORM-style code could, in principle, be pointed at a production-grade database (e.g., PostgreSQL) with minimal changes.

10. Exploratory Data Analysis

With clean, structured data available in SQLite, Day 3 focused on exploratory data analysis (EDA) to understand the underlying shape of the industry before moving to formal performance and risk metrics. SQL queries were used to aggregate data directly from the database, and results were visualised primarily using Plotly for interactivity.

10.1 NAV Trends

NAV history for each of the 40 schemes was plotted over time to identify overall growth trajectories, periods of drawdown, and relative volatility between categories. Equity-oriented schemes showed visibly higher amplitude in NAV movement compared to debt-oriented schemes, consistent with their higher-risk mandates.

10.2 AUM Analysis

Assets Under Management were analysed both in aggregate and by fund house, revealing that AUM growth is concentrated among a small number of large fund houses — a pattern consistent with the broader Indian mutual fund industry, where the top asset managers account for a disproportionate share of total AUM.

10.3 SIP Inflow Analysis

Monthly SIP inflow data was aggregated to observe the overall growth trend in systematic investment, which has become the dominant mode of retail participation in mutual funds. The analysis also broke down SIP inflows by category to identify which fund types attract the most consistent monthly investment.

10.4 Category Inflow Analysis

Net inflows and outflows were examined at the category level (equity, debt, hybrid, and others), highlighting periods where investor sentiment shifted between asset classes — useful context for interpreting subsequent performance and risk results.

10.5 Investor Transaction Analysis

Transaction-level data was analysed to understand the split between SIP, lumpsum, and redemption transactions, as well as how transaction volumes vary by investor state and age group, laying the groundwork for the Investor Analytics dashboard page.

10.6 SQL Queries and Interactive Charts

A library of parameterised SQL queries was developed against the SQLite database to support repeatable aggregation (e.g., monthly AUM by category, top 10 schemes by folio count), and the results of these queries were rendered as interactive Plotly charts, allowing hover-based inspection of exact values — a capability that static Matplotlib charts alone would not provide.

11. Performance Analytics

Day 4 of the project moved from descriptive EDA to formal, quantitative performance analytics — the core financial engineering component of the project. For each of the 40 mutual fund schemes, a consistent set of return and risk-adjusted performance metrics was calculated from the cleaned NAV history.

11.1 Metrics Computed

Metric	Definition / Purpose
Daily Returns	Day-over-day percentage change in NAV; the base series for all downstream ratio calculations
CAGR (1Y, 3Y)	Compound Annual Growth Rate over 1-year and 3-year windows, measuring annualised growth
Sharpe Ratio	Excess return over the risk-free rate per unit of total volatility (standard deviation)
Sortino Ratio	Excess return per unit of downside volatility only, penalising harmful volatility more precisely than Sharpe
Alpha	Return in excess of what would be predicted by the fund's Beta relative to its benchmark
Beta	Sensitivity of fund returns to benchmark index movements
Maximum Drawdown	Largest peak-to-trough decline in NAV over the observed period, indicating worst-case historical loss

11.2 Benchmarking Against Indices

Each scheme's returns were compared against the relevant benchmark index from the benchmark_indices dataset (e.g., NIFTY 50 for large-cap equity schemes), enabling Alpha and Beta calculations that express fund performance relative to the broader market rather than in isolation.

11.3 Fund Ranking and Scorecard

All computed metrics were consolidated into a single Fund Scorecard, and schemes were ranked primarily on risk-adjusted return (Sharpe and Sortino Ratios) rather than raw CAGR alone, reflecting the principle that a fund's return must be evaluated relative to the risk taken to achieve it.

11.4 Visual Outputs

- Benchmark comparison chart: overlaying scheme NAV growth against its benchmark index.
- Daily return distribution: histogram/density view of each scheme's daily return series, used to visually assess volatility and skew.
- Alpha-Beta report: tabular and chart-based summary of each scheme's Alpha and Beta versus its benchmark.

- CAGR report: 1-year and 3-year CAGR presented side-by-side across all 40 schemes.
- Fund Scorecard: the consolidated ranking table combining all metrics into a single reference view.

12. Risk Analytics

While Section 11 focuses on return and risk-adjusted return, a rigorous fund analysis also requires a dedicated view of downside risk. Within this project, risk analytics spans both the standard risk metrics computed in Day 4 (Maximum Drawdown, and the volatility embedded in the Sharpe/Sortino calculations) and the advanced risk metrics computed in Day 6 (Historical VaR and CVaR, detailed in Section 14).

12.1 Volatility and Drawdown

Standard deviation of daily returns was used as the base measure of volatility feeding into the Sharpe Ratio, while Maximum Drawdown was computed as the largest cumulative decline from a running peak NAV, giving a concrete, interpretable measure of the worst historical loss an investor in each scheme would have experienced.

12.2 Category-Level Risk Patterns

Risk metrics were compared across fund categories, and the results align with financial theory: small-cap and sector/thematic equity schemes displayed higher volatility and drawdown than large-cap or debt-oriented schemes. This pattern is directly reflected in the key result that the SBI Small Cap Fund recorded the highest Value-at-Risk among the funds studied.

12.3 Risk-Return Trade-off

Plotting return against risk (as implemented on the Fund Performance dashboard page, Section 13.2) makes the risk-return trade-off visually explicit, allowing a viewer to quickly identify schemes that deliver superior returns without a proportional increase in risk — the schemes of greatest interest to a risk-aware investor.

13. Dashboard Overview

Day 5 of the project translated the SQL-backed analytical layer into a four-page interactive Power BI dashboard, designed to serve three distinct audiences: industry-level stakeholders (Page 1), fund analysts and investors comparing schemes (Page 2), and behavioural/operations teams studying investor patterns (Pages 3 and 4).

13.1 Page 1 — Industry Overview



[Figure 1: Power BI Dashboard — Industry Overview]

This page provides a top-level snapshot of the mutual fund industry represented in the dataset, using KPI cards for Total AUM, Monthly SIP, Total Folios, and Total Schemes, alongside a time-series chart of the Industry AUM Trend and a breakdown of AUM by Asset Management Company.

13.2 Page 2 — Fund Performance



[Figure 2: Power BI Dashboard — Fund Performance]

The Fund Performance page is built around a Return vs Risk scatter plot, allowing quick visual identification of the risk-return trade-off described in Section 12.3, supported by a detailed Fund Table listing all computed metrics and a NAV vs Benchmark comparison chart. Slicers for Fund House, Category, and Plan allow users to narrow the view to a specific segment of interest.

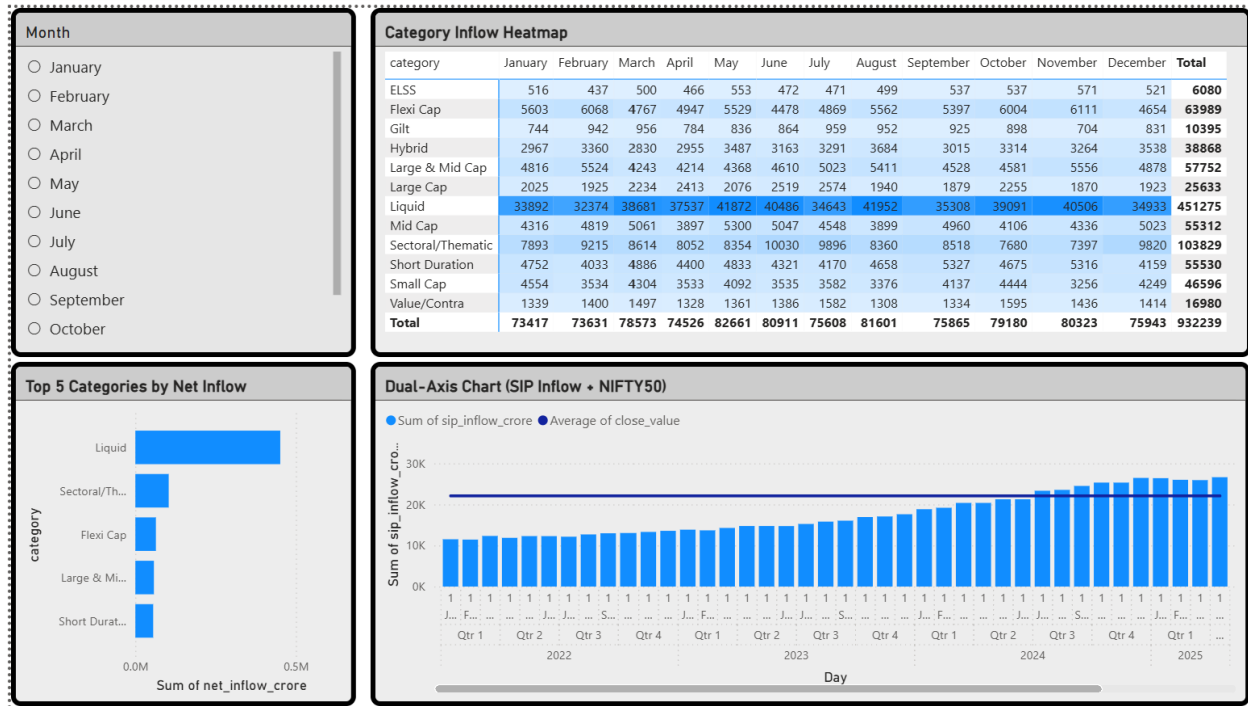
13.3 Page 3 — Investor Analytics



[Figure 3: Power BI Dashboard — Investor Analytics]

This page shifts focus from funds to investors, presenting Transactions by State on a map/bar visual, a donut chart splitting transactions into SIP, Lumpsum, and Redemption, an Age Group Analysis of investor participation, and a Monthly Transactions trend, all supported by filters for interactive exploration.

13.4 Page 4 — SIP & Market Trends



[Figure 4: Power BI Dashboard — SIP & Market Trends]

The final page presents a Category Heatmap of SIP activity, a comparison of SIP inflows against the NIFTY 50 index, a ranked view of Top Categories by inflow, and a Month Filter for temporal drill-down.

13.5 Interactivity Features

- Drill-through page: enables navigation from a summary visual (e.g., a fund house in the AUM chart) to a detailed, filtered view.
- Tooltips: custom tooltip pages/visuals that surface additional context (e.g., exact metric values) on hover without cluttering the main view.
- Interactive filters and slicers: consistent Fund House, Category, Plan, and Month filters applied across relevant pages, enabling cross-page-consistent exploration.

14. Advanced Analytics

Day 6 extended the project beyond standard performance and risk metrics into a set of advanced analytics techniques commonly used by institutional risk and product teams. This layer was designed to demonstrate depth beyond a typical academic capstone, incorporating tail-risk measurement, time-varying risk assessment, behavioural segmentation, and a lightweight recommendation system.

14.1 Historical Value-at-Risk (95%) and CVaR

Historical VaR at the 95% confidence level was computed for each scheme directly from its empirical daily return distribution, estimating the maximum expected loss over a given horizon that would not be exceeded 95% of the time. Conditional VaR (CVaR), also known as Expected Shortfall, was computed as the average loss in the worst 5% of outcomes, providing a more conservative tail-risk estimate than VaR alone. Results were exported to `var_cvar_report.csv`.

14.2 Rolling 90-Day Sharpe Ratio

Rather than relying on a single, static Sharpe Ratio computed over the full sample period, a rolling 90-day Sharpe Ratio was calculated to visualise how each scheme's risk-adjusted performance evolves over time. This reveals periods of strong or weak risk-adjusted performance that a single full-period ratio would mask, and the results were plotted in `rolling_sharpe_chart.png`.

14.3 Investor Cohort Analysis

Investors were grouped into cohorts based on their entry period (the month/quarter of their first transaction), and the behaviour of each cohort — in terms of continued SIP participation, additional lumpsum investment, and redemption — was tracked over subsequent periods, a technique adapted from cohort retention analysis used in subscription and product analytics.

14.4 SIP Continuity Analysis

SIP continuity was measured by tracking whether investors who initiated a SIP continued making contributions in subsequent months, or discontinued (lapsed) their SIP. This metric is a useful proxy for investor conviction and product stickiness, and is particularly relevant given that SIPs are the dominant channel of retail mutual fund investment in India.

14.5 Rule-Based Mutual Fund Recommender

A rule-based recommender (implemented in `recommender.py`) was built to suggest funds based on simple, explainable rules — for example, matching a user-specified risk tolerance and investment horizon to schemes with corresponding volatility and category characteristics, and prioritising schemes with strong risk-adjusted performance from the Fund Scorecard. A rule-based approach was chosen over a black-box machine learning model to keep recommendations transparent and auditable, which is particularly important in a financial advisory context.

14.6 Portfolio Concentration — Herfindahl-Hirschman Index (HHI)

Using the portfolio holdings dataset, the Herfindahl-Hirschman Index was computed for each scheme by summing the squares of individual holding weights, producing a single number that reflects how concentrated or diversified a scheme's portfolio is. Higher HHI values indicate a more concentrated portfolio (and correspondingly higher idiosyncratic risk), while lower values indicate broader diversification. Results were exported to sector_hhi.csv.

15. Key Findings

The consolidated analysis across performance, risk, and advanced analytics layers produced several clear, interpretable findings:

- The Mirae Asset Large Cap Fund recorded the highest Sharpe Ratio among all 40 schemes analysed, indicating the best risk-adjusted return per unit of total volatility in the sample.
- The Mirae Asset Large Cap Fund also recorded the highest Sortino Ratio, reinforcing that its risk-adjusted outperformance holds even when the analysis is restricted to downside volatility specifically.
- The SBI Small Cap Fund recorded the highest Value-at-Risk (95%) among the funds studied, confirming that small-cap exposure carries materially higher tail risk than large-cap exposure in this dataset — consistent with the category-level risk pattern discussed in Section 12.2.
- AUM growth and SIP inflows are concentrated among a relatively small number of large fund houses and equity-oriented categories, reflecting broader industry consolidation trends.
- SIP continuity analysis indicates that a meaningful share of investors sustain their SIP contributions over multiple months, supporting the view that systematic investment plans encourage disciplined, long-horizon investing behaviour.
- Portfolio concentration (HHI) varies meaningfully across schemes, with some funds holding significantly more concentrated portfolios than others in the same category — a factor that is not visible from return figures alone but materially affects risk.
- The full ranking of all 40 mutual funds, combining CAGR, Sharpe Ratio, Sortino Ratio, and drawdown, provides a single reference point that is more informative for decision-making than any individual metric in isolation.

16. Recommendations

Based on the findings above, the following recommendations are offered, both for prospective investors interpreting the dashboard and for future iterations of the project:

- Investors seeking capital preservation alongside reasonable growth should prioritise schemes with strong Sharpe and Sortino Ratios (such as the Mirae Asset Large Cap Fund in this dataset) over schemes with the highest raw CAGR alone.
- Investors considering small-cap schemes such as the SBI Small Cap Fund should size their allocation with awareness of the materially higher VaR observed, and consider pairing such exposure with lower-volatility categories to manage overall portfolio risk.
- Fund houses and advisory platforms could use the SIP continuity and cohort analysis to design targeted retention interventions for investor cohorts showing early signs of SIP discontinuation.
- The HHI-based concentration analysis should be incorporated into standard fund fact sheets, since two schemes with similar returns and similar Sharpe Ratios can carry very different concentration risk.
- The rule-based recommender could be extended with a feedback loop that tracks whether recommended funds subsequently met investor risk-return expectations, enabling iterative rule refinement.

17. Limitations

As with any analytics project built on a fixed dataset and a defined project timeline, several limitations should be acknowledged when interpreting the results:

- The analysis is based on the ten provided datasets and the historical period they cover; results may not generalise to market conditions outside this observed window.
- Risk metrics such as VaR and CVaR are computed using the historical simulation method, which assumes that historical return distributions are a reasonable proxy for future risk — an assumption that can break down during unprecedented market events.
- The rule-based recommender relies on explicit, human-defined rules rather than a learned model, which keeps it transparent but may miss subtler patterns that a machine-learning-based recommender could capture with sufficient data.
- Corporate actions, expense ratio changes, and scheme mergers/renames over time were not separately modelled and are assumed to be reflected consistently in the provided NAV history.
- The investor cohort and SIP continuity analyses are only as granular as the transaction dataset provided, and do not incorporate external factors such as macroeconomic events or investor-specific life events that could explain SIP discontinuation.

18. Future Scope

The current implementation provides a strong foundation that can be extended in several directions:

- Migrating the SQLite database to a cloud-hosted relational database (e.g., PostgreSQL) to support larger datasets, concurrent access, and scheduled automated refreshes.
- Automating the live NAV-fetching component into a scheduled job so the dashboard reflects near-real-time data rather than periodic manual refreshes.
- Replacing or augmenting the rule-based recommender with a machine-learning-based recommendation model trained on historical investor-fund matching outcomes.
- Extending risk analytics with parametric and Monte Carlo VaR methods alongside the current historical simulation approach, to cross-validate tail-risk estimates.
- Adding a natural-language query layer (e.g., a chatbot interface) on top of the SQLite database to allow non-technical users to ask questions about fund performance directly.
- Publishing the Power BI dashboard to the Power BI Service with row-level security, enabling role-based access for different categories of users (investors, advisors, internal analysts).

19. Conclusion

Bluestock Mutual Fund Analytics demonstrates a complete, production-style analytics workflow applied to the Indian mutual fund industry — spanning data ingestion and cleaning, relational database design, exploratory analysis, quantitative performance and risk analytics, interactive business-intelligence dashboarding, and advanced statistical and behavioural analytics. By combining standard financial metrics (CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown) with advanced techniques (VaR, CVaR, rolling risk metrics, cohort analysis, and portfolio concentration), the project goes beyond a typical descriptive dashboard to provide a genuinely diagnostic view of the mutual fund landscape.

The key results — the Mirae Asset Large Cap Fund's leadership on risk-adjusted return metrics, and the SBI Small Cap Fund's elevated tail risk — validate that the pipeline produces results consistent with financial theory and observable market behaviour, lending confidence to the broader analytical framework built over the course of the project.

Beyond its immediate results, the project served as a practical exercise in applying a Computer Science and Engineering (AI & Data Science) skill set — Python programming, SQL and database design, statistical computation, and dashboard engineering — to a real, non-trivial financial domain, and provides a solid foundation for further extension into cloud-based and machine-learning-driven analytics in the future.

20. References

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